WORKED SOLUTION

STEP III 2022, Question 8

Pure • De Moivre and Irrationality

STEP III 2022, Question 8

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2022 STEP papers (a zip file):

www.ocr.org.uk/Images/696587-2022-past-papers.zip

This question has a full guided treatment in *Beyond STEP*, Section 5.5, *Rational Angles That Aren't*: https://beyondstep.keithmcnulty.org/ch05/rational-angles.html

The key idea

De Moivre's theorem with $\cos\theta$ factored out gives $\sin k\theta$ and $\cos k\theta$ as $\cos^k\theta$ (or $\sin\theta\cos^{k-1}\theta$) times a polynomial in $\tan^2\theta = \sec^2\theta - 1$. When $\sec^2\theta - 1$ is an *even integer*, that polynomial is an integer whose parity we can read off from its first term. A rational angle in degrees would make some $\sin k\theta$ vanish, and the parity argument shows that this can never happen.

Solution

Part (i)

Assume $\cos\theta \neq 0$, so that $\sec\theta$ exists. By De Moivre's theorem,

$$\cos k\theta + \mathrm{i}\sin k\theta = (\cos\theta + \mathrm{i}\sin\theta)^k = \cos^k\theta\,(1+\mathrm{i}t)^k, \qquad t = \tan\theta.$$

Expanding binomially, $(1+\mathrm{i}t)^k = \sum_{j=0}^{k}\binom{k}{j}\mathrm{i}^j t^j$. The odd-$j$ terms are imaginary, with $\mathrm{i}^{2r+1} = (-1)^r\mathrm{i}$, and the even-$j$ terms are real, with $\mathrm{i}^{2r} = (-1)^r$. Taking imaginary parts,

$$\begin{aligned}\sin k\theta &= \cos^k\theta\left(\binom{k}{1}t - \binom{k}{3}t^3 + \binom{k}{5}t^5 - \cdots\right)\\ &= \cos^k\theta\,\tan\theta\left(k - \binom{k}{3}t^2 + \binom{k}{5}t^4 - \cdots\right).\end{aligned}$$

Since $\cos^k\theta\tan\theta = \sin\theta\cos^{k-1}\theta$ and $t^2 = \tan^2\theta = \sec^2\theta - 1$, this is

$$\sin k\theta = \sin\theta\cos^{k-1}\theta\left(k - \binom{k}{3}(\sec^2\theta - 1) + \binom{k}{5}(\sec^2\theta - 1)^2 - \cdots\right),$$

as required. The bracket stops of its own accord, because $\binom{k}{j} = 0$ for $j > k$. Taking real parts in the same way,

$$\cos k\theta = \cos^k\theta \left(1 - \binom{k}{2}(\sec^2\theta - 1) + \binom{k}{4}(\sec^2\theta - 1)^2 - \cdots\right).$$

Part (ii)

Here $\cos\theta = 1/a$ with $a \geq 3$, so $0^\circ < \theta < 90^\circ$. In particular $\sin\theta \neq 0$ and $\cos\theta \neq 0$. Also $\sec^2\theta - 1 = a^2 - 1$, which is an *even* integer because a is odd. Write $X = a^2 - 1$.

Suppose k is the least positive integer with $\sin k\theta = 0$. Then $k \neq 1$, since $\sin\theta \neq 0$, so $k > 1$ and part (i) applies:

$$0 = \sin k\theta = \sin\theta\cos^{k-1}\theta \left(k - \binom{k}{3}X + \binom{k}{5}X^2 - \cdots\right).$$

The factor $\sin\theta\cos^{k-1}\theta$ is non-zero, so the bracket is zero. Every term after the first contains at least one factor of the even integer X, and binomial coefficients are integers, so

$$0 = k - \binom{k}{3}X + \binom{k}{5}X^2 - \cdots \equiv k \pmod 2.$$

Hence k is even. Now

$$0 = \sin k\theta = 2\sin\left(\tfrac{1}{2}k\theta\right)\cos\left(\tfrac{1}{2}k\theta\right).$$

But $\frac{1}{2}k$ is an integer with $0 < \frac{1}{2}k < k$, so by the minimality of k, $\sin(\frac{1}{2}k\theta) \neq 0$. Therefore $\cos(\frac{1}{2}k\theta) = 0$, as required.

Ask yourself Where does the assumption that θ is rational come in?

It is what produces a k to start with. Suppose $\theta = p/q$ with p, q positive integers (recall θ is measured in degrees). Then $180q\,\theta = 180p$, so $\sin(180q\,\theta) = 0$. So there *is* a positive integer m with $\sin m\theta = 0$, and hence a least one; call it k. It satisfies the hypotheses above, so k is even and, writing $n = \frac{1}{2}k$, we have $\cos n\theta = 0$.

We now show that $\cos n\theta$ cannot be zero. If $n = 1$, then $\cos n\theta = 1/a \neq 0$. If $n > 1$, the cosine formula from part (i) gives

$$\cos n\theta = \frac{1}{a^n}\left(1 - \binom{n}{2}X + \binom{n}{4}X^2 - \cdots\right).$$

Again every term after the first is even, so the bracket is an odd integer, and in particular non-zero. So $\cos n\theta \neq 0$, a contradiction. Hence θ is irrational.

Part (iii)

Now $\cot\phi = 1/b$, so $\tan\phi = b$ with $b \geq 2$, and $0^\circ < \phi < 90^\circ$. Again $\sin\phi \neq 0$ and $\cos\phi \neq 0$, and this time

$$\sec^2\phi - 1 = \tan^2\phi = b^2,$$

which is an *even* integer because b is even. So exactly the argument of part (ii) runs with $X = b^2$ in place of $a^2 - 1$. In full:

Suppose $\phi = p/q$ with p, q positive integers. Then $\sin(180q\,\phi) = 0$, so there is a least positive integer k with $\sin k\phi = 0$, and $k > 1$ as $\sin\phi \neq 0$. By part (i), since $\sin\phi\cos^{k-1}\phi \neq 0$,

$$0 = k - \binom{k}{3}b^2 + \binom{k}{5}b^4 - \cdots \equiv k \pmod 2,$$

so k is even. Then $0 = \sin k\phi = 2\sin(\tfrac{1}{2}k\phi)\cos(\tfrac{1}{2}k\phi)$ and, by minimality, $\sin(\tfrac{1}{2}k\phi) \neq 0$, so $\cos n\phi = 0$ where $n = \tfrac{1}{2}k$. If $n = 1$ this says $\cos\phi = 0$, which is false. If $n > 1$,

$$\cos n\phi = \cos^n\phi\left(1 - \binom{n}{2}b^2 + \binom{n}{4}b^4 - \cdots\right),$$

where $\cos^n\phi \neq 0$ and the bracket is odd, so non-zero. This contradiction shows that ϕ is irrational.

Checks

- Try $k = 2$ and $k = 3$ in part (i): the formulas give $\sin 2\theta = 2\sin\theta\cos\theta$, $\cos 2\theta = \cos^2\theta - \sin^2\theta$ and $\sin 3\theta = \sin\theta(3\cos^2\theta - \sin^2\theta) = 3\sin\theta - 4\sin^3\theta$. Both formulas also agree with direct evaluation at random angles, for k up to 15.
- The parity conditions matter. With $a = 2$ (even) we get $\theta = 60°$, and with $b = 1$ (odd) we get $\phi = 45°$, both rational. In each case $X = 3$ or 1 is odd, so the argument breaks down exactly where it should.
- For $a = 3$, $\theta \approx 70.5288°$, and for $b = 2$, $\phi \approx 63.4349°$. A computer check confirms that $\sin k\theta$ and $\sin k\phi$ stay well away from zero for every $k < 2000$, for every odd a and even b below 30.

Common slips

- Forgetting to say why $\sin\theta\cos^{k-1}\theta \neq 0$ before dividing it out.
- Not explaining where k comes from: the minimal k exists only because a rational θ makes $\sin(180q\,\theta) = 0$.
- Concluding $\cos(\tfrac{1}{2}k\theta) = 0$ without using the minimality of k to rule out $\sin(\tfrac{1}{2}k\theta) = 0$.

Beyond STEP

The brackets here are, up to the factors in front, the *Chebyshev polynomials*, and the parity trick is a first taste of reducing integer identities modulo a prime. A famous relative is *Niven's theorem*: the only rational multiples θ of $180°$ for which $\cos\theta$ is rational are those with $\cos\theta \in \{0, \pm\frac{1}{2}, \pm 1\}$. It is proved by showing that $2\cos\theta$ is an algebraic integer.

Davenport's *The Higher Arithmetic* is a good place to build up the number theory behind arguments like this.

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